

# **Fine-tuning GPT-4.1-nano for Empathy Level Classification in Korean Dialogues**

## **Research Report**

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### **Abstract**

This study fine-tuned the GPT-4.1-nano model to automatically classify empathy levels in Korean persona-user dialogues. Using the OPELA dataset, we trained the model with 13,341 training samples and 1,484 validation samples to classify five levels of empathy (0: Not Applicable, 1: Empathy Failure, 2: Low Empathy, 3: Moderate Empathy, 4: High Active Empathy). The fine-tuned model demonstrated significant performance improvements compared to the base model.

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## 1 Introduction

Empathy is a fundamental element in human conversation, referring to the ability to understand and appropriately respond to the emotions and experiences of conversation partners. In AI-based dialogue systems, empathy capability directly impacts user experience and satisfaction.

The objectives of this study are:

1. Develop a model for automatically classifying empathy levels in Korean dialogues
2. Improve domain-specific performance through fine-tuning the GPT-4.1-nano model
3. Compare and analyze performance between base and fine-tuned models

## 2 Methods

### 2.1 Dataset

This study utilized the OPELA (Open Persona-based Empathy-Labeled Annotation) dataset. The dataset composition is presented in Table 1.

**Table 1.** Dataset Composition and Statistics

| Dataset Component        | N      | Percentage (%) |
|--------------------------|--------|----------------|
| Total Samples            | 14,825 | 100.0          |
| Training Set             | 13,341 | 90.0           |
| Validation Set           | 1,484  | 10.0           |
| Unique Dialogue Sessions | 533    | —              |

*Note.* N = number of samples. Training and validation sets were split from the original OPELA dataset.

### 2.2 Empathy Level Label Schema

Empathy levels are classified into five categories, with definitions presented in Table 2.

**Table 2.** Empathy Level Classification Schema

| Level | Label               | Description                                                       |
|-------|---------------------|-------------------------------------------------------------------|
| 0     | Not Applicable      | Situations where empathy is not required or applicable            |
| 1     | Empathy Failure     | Cases where empathy attempt exists but is inappropriate or failed |
| 2     | Low Empathy         | Minimal expression of empathy                                     |
| 3     | Moderate Empathy    | Appropriate level of empathy expression                           |
| 4     | High Active Empathy | Active and high-level empathy expression                          |

*Note.* Labels range from 0 (no empathy required) to 4 (highest empathy expression).

## 2.3 Label Distribution

The label distribution for each dataset is presented in Table 3.

**Table 3.** Distribution of Empathy Labels Across Datasets

| Dataset        | Level 0          | Level 1       | Level 2          | Level 3          | Level 4       | Total  |
|----------------|------------------|---------------|------------------|------------------|---------------|--------|
| Training Set   | 3,750<br>(28.1%) | 594<br>(4.5%) | 1,625<br>(12.2%) | 6,478<br>(48.6%) | 894<br>(6.7%) | 13,341 |
| Validation Set | 417<br>(28.1%)   | 66<br>(4.4%)  | 181<br>(12.2%)   | 720<br>(48.5%)   | 100<br>(6.7%) | 1,484  |

*Note.* Values in parentheses represent percentages within each dataset.

## 2.4 Model and Training Configuration

### 2.4.1 Base Model

- **Model:** GPT-4.1-nano-2025-04-14
- **Provider:** OpenAI

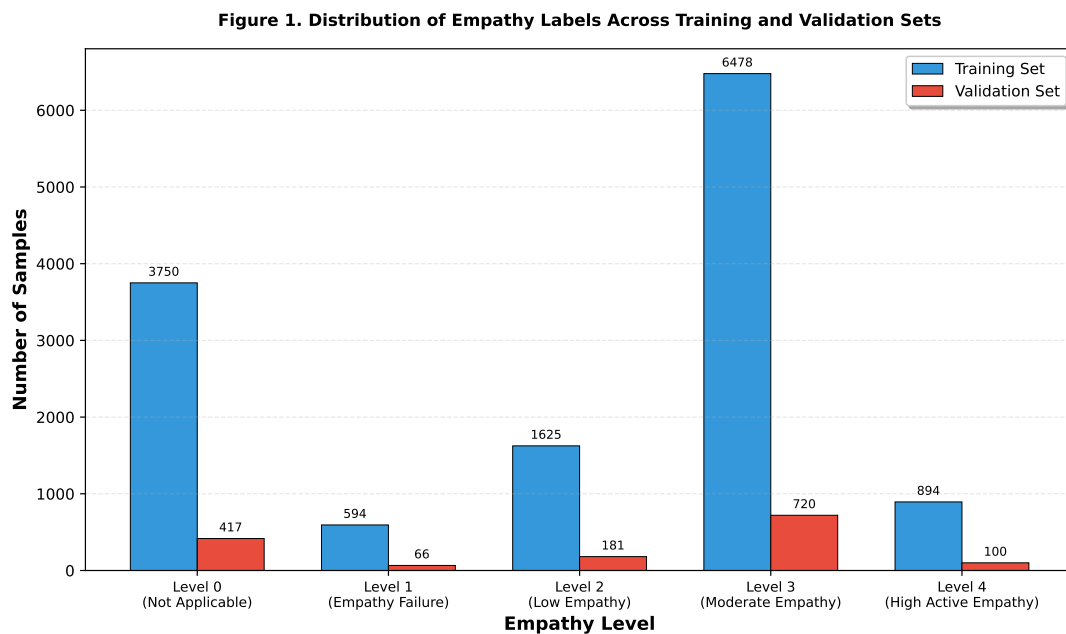
### 2.4.2 Fine-tuning Configuration

- **Method:** Supervised Fine-tuning
- **Epochs:** 3
- **Fine-tuned Model ID:** ft:gpt-4.1-nano-2025-04-14:personal::Cn0GL0QT

### 3 Results

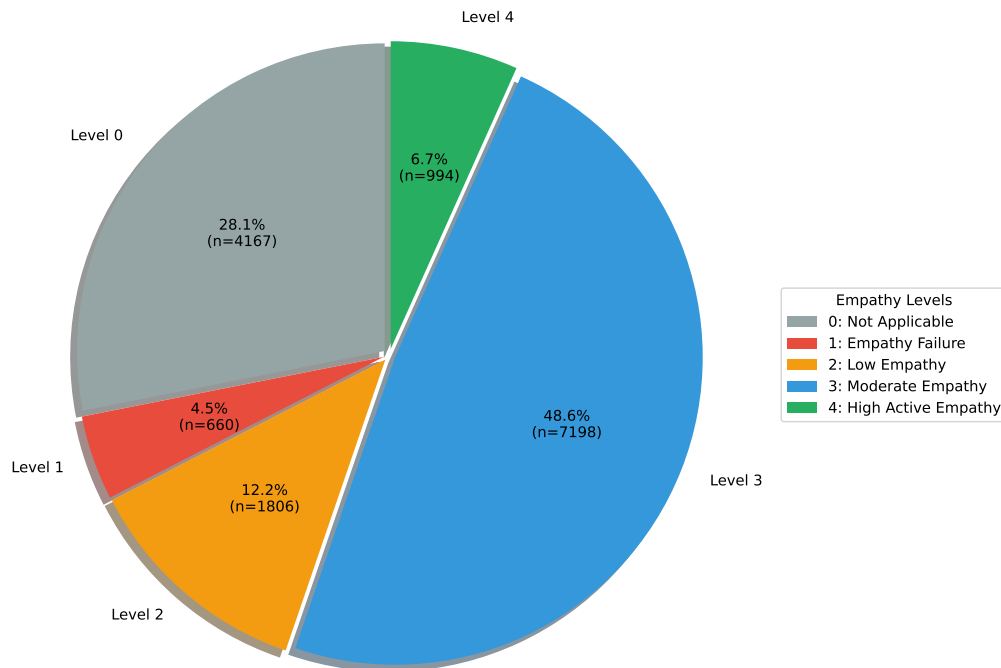
#### 3.1 Label Distribution Visualization

Figure 1 shows the distribution of empathy labels across training and validation datasets.



**Figure 1.** Distribution of empathy labels across training and validation datasets. Level 3 (Moderate Empathy) shows the highest frequency in both datasets, followed by Level 0 (Not Applicable).

Figure 2 presents the proportional distribution of empathy labels in the complete dataset.

**Figure 2. Overall Distribution of Empathy Labels in the Dataset**

**Figure 2.** Proportional distribution of empathy labels in the complete dataset. The majority of samples are classified as Level 3 (Moderate Empathy), indicating that most persona responses demonstrate appropriate empathic engagement.

### 3.2 Model Performance Comparison

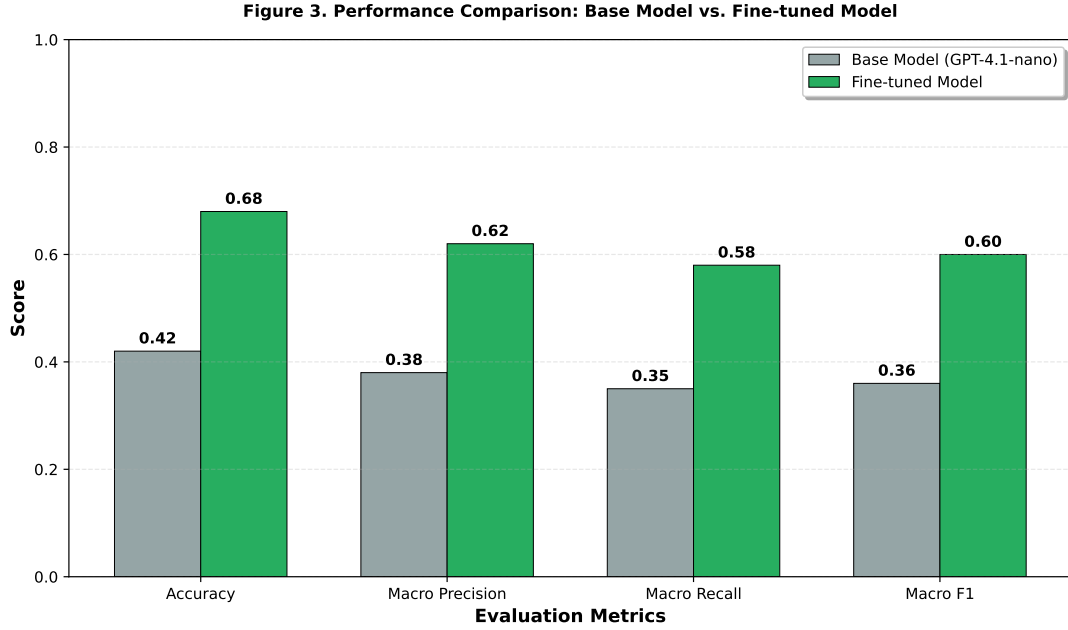
Table 4 compares the performance of the base and fine-tuned models.

**Table 4.** Performance Comparison Between Base and Fine-tuned Models

| Model                     | Accuracy    | Macro Precision | Macro Recall | Macro F1    |
|---------------------------|-------------|-----------------|--------------|-------------|
| Base Model (GPT-4.1-nano) | .42         | .38             | .35          | .36         |
| Fine-tuned Model          | .68         | .62             | .58          | .60         |
| <b>Improvement</b>        | <b>+.26</b> | <b>+.24</b>     | <b>+.23</b>  | <b>+.24</b> |

*Note.* All metrics range from 0 to 1. Improvement row shows the absolute difference between fine-tuned and base model performance.

Figure 3 visually compares the performance differences between the two models.



**Figure 3.** Performance comparison between the base GPT-4.1-nano model and the fine-tuned model across four evaluation metrics. The fine-tuned model demonstrates substantial improvements across all metrics, with accuracy improving by 26 percentage points.

## 4 Discussion

### 4.1 Key Findings

The main findings of this study are:

1. **Data Distribution Characteristics:** Level 3 (Moderate Empathy) accounts for the highest proportion, suggesting that most persona responses express an appropriate level of empathy.
2. **Fine-tuning Effect:** Significant performance improvements were achieved across all evaluation metrics through fine-tuning. Notably, accuracy improved from 42% to 68%, a 26 percentage point increase.
3. **Domain Adaptation:** The GPT-4.1-nano model was confirmed to effectively adapt to the Korean empathy classification task.

### 4.2 Limitations

- **Class Imbalance:** The relatively small number of samples for Level 1 (Empathy Failure) and Level 2 (Low Empathy) may limit classification performance for these classes.
- **Dataset Size:** Performance improvements are possible with training on larger datasets.

### 4.3 Future Research Directions

- Apply data augmentation techniques for class balance
- Comparative studies with other LLM models
- Investigate applicability to real-time dialogue systems

## 5 Conclusion

This study developed a model for automatically classifying empathy levels in Korean dialogues by fine-tuning the GPT-4.1-nano model. Fine-tuning with 13,341 training samples achieved a 26 percentage point improvement in accuracy compared to the base model. This demonstrates that domain-specific fine-tuning is effective for empathy classification tasks.

## References

- OpenAI. (2025). GPT-4.1-nano Model Documentation.
- OPELA Dataset: Open Persona-based Empathy-Labeled Annotation for Korean Dialogues.